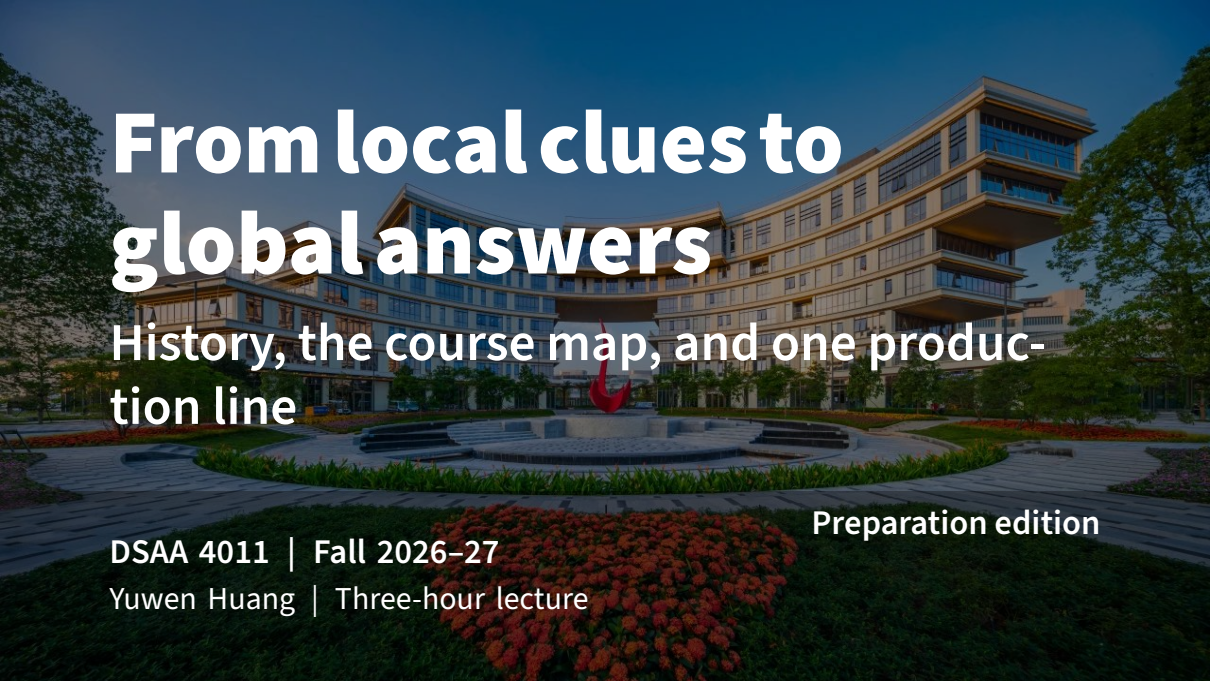




From local clues to global answers

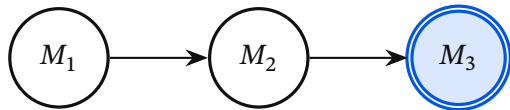
History, the course map, and one production line

DSAA 4011 | Fall 2026–27

Yuwen Huang | Three-hour lecture

Preparation edition

A production line raises one question



unobserved

target

alarm observed

**Machine 3 reports an alarm.
What should we conclude
about Machine 2?**

Write before discussion.

observation

local relations

conclusion

Why this question is difficult

Missing information

Machines 1 and 2 are not directly observed.

Dependence

A fault at one machine can change the state of its neighbor.

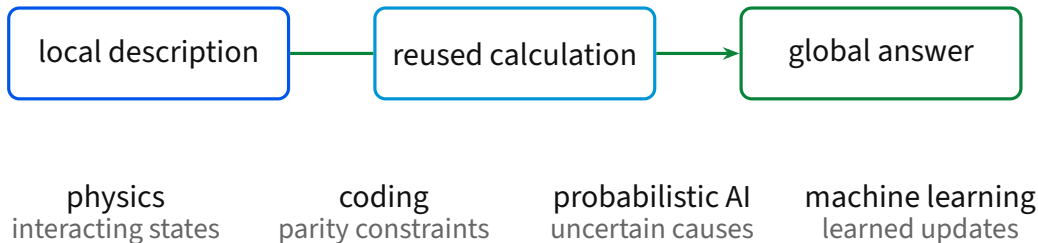
Ambiguity

Several unseen states can explain the same alarm.

Required result

We need a probability for each possible state of Machine 2.

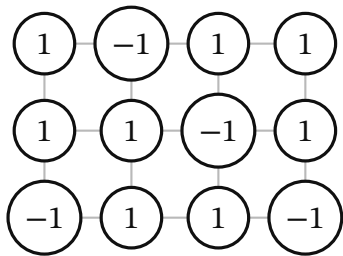
One idea appears across many fields



A short history

The applications changed; the local-to-global problem remained.

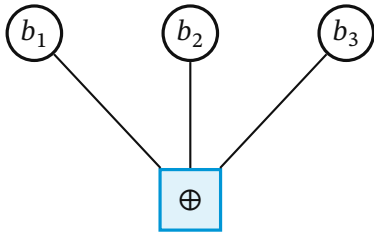
1920s: neighboring states interact



Physicists described a large system through interactions between nearby states.

Lesson. A global pattern can arise from many small local relations.

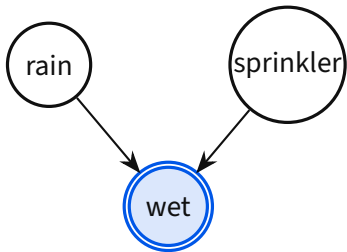
1950s: local constraints recover data



Parity constraints relate small groups of transmitted bits. A decoder combines them to recover missing or corrupted bits.

Lesson. Local calculations can be reused instead of checking every codeword.

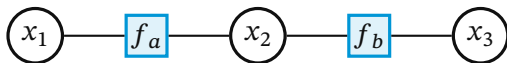
1980s: graphs organize uncertainty



Probabilistic graphical models separated uncertain causes, observations, and questions.

The graph shows dependence. Numerical probabilities and a query are still required.

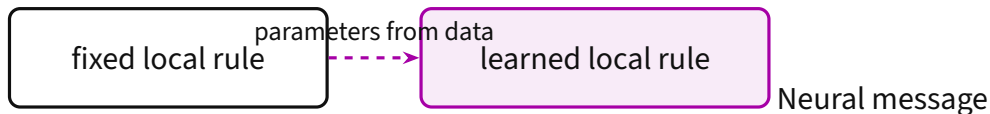
Factor graphs give a common picture



A factor graph records which variables each local function uses.

Lesson. The same representation supports many exact and approximate calculations.

Today: some update rules are learned



passing, multiscale networks, diffusion, and attention reuse local or repeated computations.

A similar computation pattern does not give these models the same probabilistic meaning.

The course develops one question

Weeks 1-2

Represent local relations and compute exact answers on trees.

Weeks 3-7

Measure uncertainty, learn parameters and structure, and approximate hard calculations.

Weeks 9-11

Compare fixed calculations with learned structured computation.

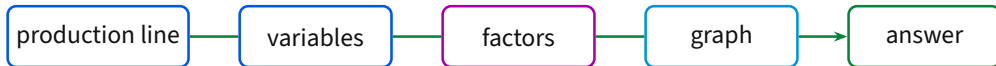
Weeks 12-14

Study repeated denoising, attention, and language-model prediction.

What remains constant across these topics?

Write before discussion.

Week 1 builds the first exact answer



Define first. Every variable, value, function, and output.

Calculate second. Multiply local weights, sum unseen cases, and normalize.

Check last. Compare the result with an independent calculation.

One production-line model

Define every object, then calculate one conditional probability.

Describe one small system

Question

A probability question needs declared variables and values.

Main idea

Use three binary variables and list their assignments.

Evidence

Translate a verbal description into notation.

Define the variables

Let $V = \{1, 2, 3\}$ be the set of machine indices.

For each $i \in V$, define the binary variable

$$x_i \in \mathcal{X}_i = \{0, 1\},$$

where 0 means normal and 1 means alert. The symbol x_i records the state of Machine i .

A **complete assignment** is $x = (x_1, x_2, x_3)$. For example, $(0, 1, 1)$ means that Machines 2 and 3 are alert.

Variable, value, or assignment?

x_2

a variable

variable

$x_2 = 1$

a value statement

declared value

$(0, 1, 1)$

a complete
assignment

all variables fixed

List all eight assignments

Three binary variables give $2^3 = 8$ complete assignments.

(0, 0, 0) (0, 0, 1) (0, 1, 0) (0, 1, 1)

(1, 0, 0) (1, 0, 1) (1, 1, 0) (1, 1, 1)

Each tuple fixes one value for every variable.

Apply the observation $x_3 = 1$

(0, 0, 1)

(0, 1, 1)

(1, 0, 1)

(1, 1, 1)

Why

were the other four assignments removed?

Write before discussion.

Why not start with eight probabilities?

A full table assigns one number to every complete assignment. For n binary variables, it needs 2^n entries.

The system has a simpler description: Machine 1 interacts locally with Machine 2, and Machine 2 interacts locally with Machine 3.

Local structure lets us describe these two interactions separately.

Local checkpoint

Complete this checkpoint after the section.

Conclusion

Condition

**Common
error**

Next step

2. Build local factors

Each neighboring pair favors some joint values more than others.

Build local factors roadmap

Question

Each neighboring pair favors some joint values more than others.

Main idea

Use one nonnegative table for each local interaction.

Evidence

Read a factor value from its declared arguments and table axes.

Define the first factor

The factor f_{12} uses variables x_1 and x_2 :

$$f_{12} : \mathcal{X}_1 \times \mathcal{X}_2 \rightarrow \mathbb{R}_{\geq 0}.$$

	$x_2 = 0$	$x_2 = 1$
$x_1 = 0$	4	1
$x_1 = 1$	2	5

For example, $f_{12}(1, 0) = 2$. Larger entries indicate machine-state pairs that fit the historical record better. The table gives **weights**, not probabilities.

Read a factor table

	$x_2 = 0$	$x_2 = 1$
$x_1 = 0$	1	2
$x_1 = 1$	4	1

What is $f_{12}(1, 0)$?

Write before discussion.

Define the second factor

The factor f_{23} uses variables x_2 and x_3 :

$$f_{23} : \mathcal{X}_2 \times \mathcal{X}_3 \rightarrow \mathbb{R}_{\geq 0}.$$

	$x_3 = 0$	$x_3 = 1$
$x_2 = 0$	3	1
$x_2 = 1$	1	4

For example, $f_{23}(0, 1) = 1$. The row now represents x_2 , not x_1 .

Combine one complete assignment

For a complete assignment $x = (x_1, x_2, x_3)$, define its unnormalized weight by

$$w(x) = f_{12}(x_1, x_2)f_{23}(x_2, x_3).$$

For $x = (1, 0, 1)$, the factors use the entries $f_{12}(1, 0)$ and $f_{23}(0, 1)$. **Compute $w(1, 0, 1)$.**

Write before discussion.

Multiplication combines local weights that hold for the same assignment.

A factor is not a probability table

The entries of f_{12} sum to 12, and the entries of f_{23} sum to 9. Neither table is a probability distribution.

The factors specify **relative compatibility**. We normalize only after combining the factors and selecting the assignments relevant to the query.

Common error. Normalizing each factor separately generally changes the model.

Local checkpoint

Complete this checkpoint after the section.

Conclusion

Condition

**Common
error**

Next step

3. Draw the factor graph

A formula is precise, but a graph makes its local structure visible.

Draw the factor graph roadmap

Question

A formula is precise, but a graph makes its local structure visible.

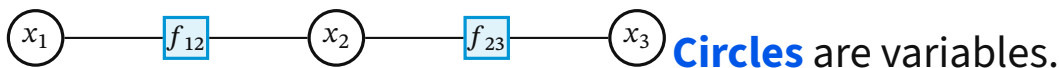
Main idea

Use circles for variables, squares for factors, and edges for factor arguments.

Evidence

Recover every factor argument from the graph.

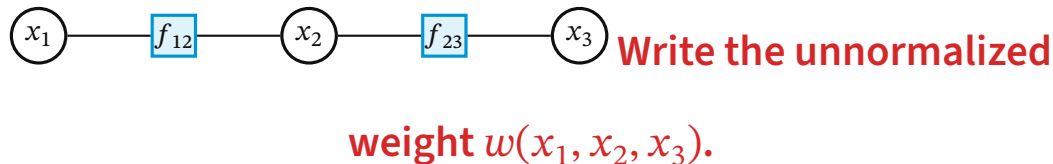
Draw the concrete graph



Circles are variables.

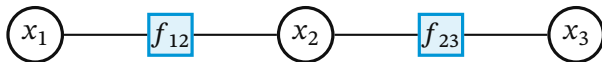
Squares are factors. An edge means that the factor uses the connected variable as an argument.

From graph to formula



Write before discussion.

Read the graph from its edges



f_{12} uses (x_1, x_2) .

f_{23} uses (x_2, x_3) .

Removing one edge changes the model

What changes if the edge between x_2 and f_{23} is removed?

Write before discussion.

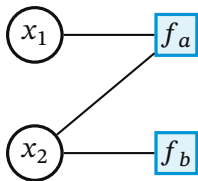
Define a factor graph

Let V index variable nodes and F index factor nodes. For $a \in F$, let $S_a \subseteq V$ list the variables used by f_a .

$$G = (V \cup F, E), \quad E = \{(i, a) : i \in S_a\}.$$

An edge (i, a) means that x_i is an argument of f_a .

What bipartite means



Every edge joins a **variable node** to a **factor**

node. The graph records arguments, not arithmetic operations.

Graph and model have different roles

Graph

Which variables does each factor use?

Model

$$w(x) = \prod_{a \in F} f_a(x_{S_a})$$

How do local values combine?

The query chooses what to retain

$$\text{retain } x_2 \quad \Leftarrow \quad \sum_{x_1} w(x_1, x_2, x_3 = 1)$$

Summation removes the hidden variable x_1 . A different algebra must be declared explicitly.

Local checkpoint

Complete this checkpoint after the section.

Conclusion

Condition

**Common
error**

Next step

4. Answer one probability query

Several complete assignments agree with the same observed and queried values.

Answer one probability query roadmap

Question

Several complete assignments agree with the same observed and queried values.

Main idea

Multiply factor values within each assignment, add compatible assignments, then normalize.

Evidence

Produce the two probabilities and verify that they sum to one.

Define the probability notation

After normalization, p denotes the probability distribution defined by the factor weights.

The expression

$$p(x_2 = j \mid x_3 = 1)$$

means the probability that Machine 2 has state j , given the observation that Machine 3 is alert. Here $j \in \{0, 1\}$.

The vertical bar means “given.” It does not mean division.

State the observation and query

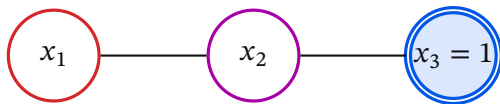
Observation. We observe $x_3 = 1$.

Query. Compute the conditional distribution of x_2 :

$$\left[p(x_2 = 0 \mid x_3 = 1), p(x_2 = 1 \mid x_3 = 1) \right].$$

The value of x_1 remains unobserved. We must include both $x_1 = 0$ and $x_1 = 1$.

Visible, retained, and eliminated



hidden: sum out

query: retain

observed: fix

sum over x_1

retain x_2

set $x_3 = 1$

Weight for the answer $x_2 = 0$

With $x_2 = 0$ and $x_3 = 1$, two assignments remain:

Complete both products, then add them to obtain r_0 .

Write before discussion.

Weight for the answer $x_2 = 1$

With $x_2 = 1$ and $x_3 = 1$, two assignments remain:

Complete both products, then add them to obtain r_1 .

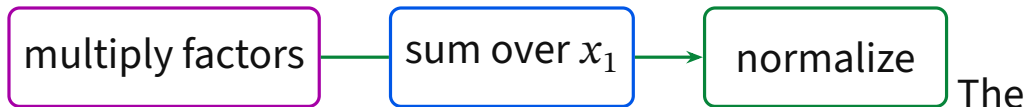
Write before discussion.

Normalize the two weights

Normalize r_0 and r_1 . Then state two checks on the result.

Write before discussion.

Three operations answer the query



first two operations form **sum-product**; normalization turns the retained weights into probabilities.

Write the same calculation compactly

Let $j \in \mathcal{X}_2$ be a possible value of x_2 . Define

$$r_j = \sum_{i \in \mathcal{X}_1} f_{12}(i, j) f_{23}(j, 1).$$

Then

$$p(x_2 = j \mid x_3 = 1) = \frac{r_j}{\sum_{j' \in \mathcal{X}_2} r_{j'}}.$$

Every symbol in these equations has already appeared in the concrete calculation.

A plausible wrong calculation

An AI assistant proposes

$$r_j = \sum_i [f_{12}(i, j) + f_{23}(j, 1)].$$

What rule did it change?

Write before discussion.

Local checkpoint

Complete this checkpoint after the section.

Conclusion

Condition

**Common
error**

Next step

5. Next: local messages

Enumeration is correct but repeats the same partial sums.

Next: local messages roadmap

Question

Enumeration is correct but repeats the same partial sums.

Main idea

Group reusable local calculations and pass their results across the graph.

Evidence

Identify the repeated quantity that Week 2 will call a message.

Find the repeated calculation

For each value j of x_2 , the calculation first forms

$$m_{1 \rightarrow 2}(j) = \sum_{i \in \mathcal{X}_1} f_{12}(i, j).$$

It then combines this result with the observed-side factor:

$$r_j = m_{1 \rightarrow 2}(j) f_{23}(j, 1).$$

The symbol $m_{1 \rightarrow 2}$ names a reusable table; Week 2 will derive the general rule.

Why local reuse matters

Direct enumeration visits every complete assignment. A model with n binary variables may have 2^n assignments.

On a tree, local messages summarize one part of the graph before the rest of the graph uses it.

The result remains exact, but the computation avoids repeating the same partial sums.

AI can implement the wrong rule

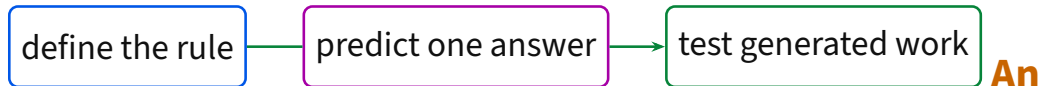
Model errors

- add instead of multiply
- use the wrong factor axis

Query errors

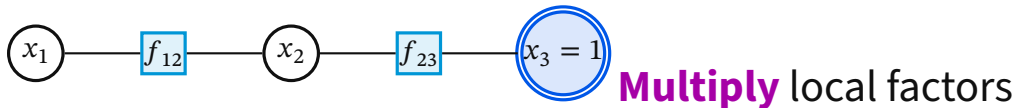
- retain the wrong variable
- normalize over the wrong dimension

Your evidence must be independent



**AI explanation cannot independently verify
AI-generated code.**

Week 1 in one picture



for one assignment.

Sum over the hidden value x_1 .

Normalize across the two values of x_2 .

Exit check

In one sentence, distinguish the graph from the sum--product calculation.

Write before discussion.

Next week. Derive exact local message rules for any tree-structured factor graph.